RUN_EVIDENCE: OmniTrainer multimodal moderation

1. Test output: `uv run pytest tests/ -vv --ignore=tests/test_env_setup.py`

```
============================= test session starts ==============================
platform linux -- Python 3.12.12, pytest-8.4.2, pluggy-1.6.0 -- /voc/work/code/project/starter/udacity-generative-ai-section-3-project/.venv/bin/python
cachedir: .pytest_cache
rootdir: /voc/work/code/project/starter/udacity-generative-ai-section-3-project
configfile: pyproject.toml
plugins: anyio-4.11.0, logfire-4.13.2, asyncio-1.2.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collecting ... collected 45 items

tests/test_audio_agent.py::test_moderate_audio_exists PASSED             [  2%]
tests/test_audio_agent.py::test_moderate_audio_returns_audio_moderation_result PASSED [  4%]
tests/test_audio_agent.py::test_moderate_audio_has_required_fields PASSED [  6%]
tests/test_audio_agent.py::test_moderate_audio_rationale_not_empty PASSED [  8%]
tests/test_audio_agent.py::test_moderate_audio_transcription_not_empty PASSED [ 11%]
tests/test_gradio_app.py::test_create_chat_interface_returns_blocks PASSED [ 13%]
tests/test_gradio_app.py::test_chatbot_uses_messages_type PASSED         [ 15%]
tests/test_gradio_app.py::test_chatbot_has_copy_button PASSED            [ 17%]
tests/test_gradio_app.py::test_demo_has_chatbot_component PASSED         [ 20%]
tests/test_gradio_app.py::test_demo_has_multimodal_textbox PASSED        [ 22%]
tests/test_gradio_app.py::test_multimodal_textbox_accepts_multiple_files PASSED [ 24%]
tests/test_gradio_app.py::test_multimodal_textbox_accepts_correct_file_types PASSED [ 26%]
tests/test_gradio_app.py::test_chat_with_gemini_calls_moderation PASSED  [ 28%]
tests/test_gradio_app.py::test_chat_with_gemini_blocks_flagged_content PASSED [ 31%]
tests/test_image_agent.py::test_moderate_image_exists PASSED             [ 33%]
tests/test_image_agent.py::test_moderate_image_returns_image_moderation_result PASSED [ 35%]
tests/test_image_agent.py::test_moderate_image_has_required_fields PASSED [ 37%]
tests/test_image_agent.py::test_moderate_image_rationale_not_empty PASSED [ 40%]
tests/test_moderation_result.py::TestModerationResult::test_has_rationale_field PASSED [ 42%]
tests/test_moderation_result.py::TestModerationResult::test_rationale_is_required PASSED [ 44%]
tests/test_moderation_result.py::TestModerationResult::test_is_pydantic_model PASSED [ 46%]
tests/test_moderation_result.py::TestTextModerationResult::test_has_all_required_fields PASSED [ 48%]
tests/test_moderation_result.py::TestTextModerationResult::test_field_types PASSED [ 51%]
tests/test_moderation_result.py::TestTextModerationResult::test_inherits_from_moderation_result PASSED [ 53%]
tests/test_moderation_result.py::TestTextModerationResult::test_all_fields_are_required PASSED [ 55%]
tests/test_moderation_result.py::TestImageModerationResult::test_has_all_required_fields PASSED [ 57%]
tests/test_moderation_result.py::TestImageModerationResult::test_field_types PASSED [ 60%]
tests/test_moderation_result.py::TestImageModerationResult::test_inherits_from_moderation_result PASSED [ 62%]
tests/test_moderation_result.py::TestImageModerationResult::test_all_fields_are_required PASSED [ 64%]
tests/test_moderation_result.py::TestVideoModerationResult::test_has_all_required_fields PASSED [ 66%]
tests/test_moderation_result.py::TestVideoModerationResult::test_field_types PASSED [ 68%]
tests/test_moderation_result.py::TestVideoModerationResult::test_inherits_from_moderation_result PASSED [ 71%]
tests/test_moderation_result.py::TestVideoModerationResult::test_all_fields_are_required PASSED [ 73%]
tests/test_moderation_result.py::TestAudioModerationResult::test_has_all_required_fields PASSED [ 75%]
tests/test_moderation_result.py::TestAudioModerationResult::test_field_types PASSED [ 77%]
tests/test_moderation_result.py::TestAudioModerationResult::test_inherits_from_moderation_result PASSED [ 80%]
tests/test_moderation_result.py::TestAudioModerationResult::test_all_fields_are_required PASSED [ 82%]
tests/test_text_agent.py::test_moderate_text_exists PASSED               [ 84%]
tests/test_text_agent.py::test_moderate_text_returns_text_moderation_result PASSED [ 86%]
tests/test_text_agent.py::test_moderate_text_has_required_fields PASSED  [ 88%]
tests/test_text_agent.py::test_moderate_text_rationale_not_empty PASSED  [ 91%]
tests/test_video_agent.py::test_moderate_video_exists PASSED             [ 93%]
tests/test_video_agent.py::test_moderate_video_returns_video_moderation_result PASSED [ 95%]
tests/test_video_agent.py::test_moderate_video_has_required_fields PASSED [ 97%]
tests/test_video_agent.py::test_moderate_video_rationale_not_empty PASSED [100%]

============================== 45 passed in 13.33s ==============================
```

2. Evaluation output: `uv run evals/text/test_cases.py`

A score below 100% is expected: both the moderation agent and the LLM judge are language models.

```
Evaluating run_text_moderation ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 100% 0:00:00
                                                    Evaluation Summary: run_text_moderation
```

Case ID	Inputs	Outputs	Assertions
professional_text	[TextInput(text_file='/voc/work/code/project/starter/udacity-generative-ai-section-3-project/evals/test_data/professional_text.txt')]	rationale="The message is friendly and professional. It clearly states the company's willingness to help, provides guidance on how to proceed, and offers further assistance. No PII was detected." contains_pii=False is_unfriendly=False is_unprofessional=False	✔✔✔✔
text_with_pii	[TextInput(text_file='/voc/work/code/project/starter/udacity-generative-ai-section-3-project/evals/test_data/text_with_pii.txt')]	rationale='The message contains the customer' contains_pii=True is_unfriendly=False is_unprofessional=False	✔✗✔✔
unfriendly_text	[TextInput(text_file='/voc/work/code/project/starter/udacity-generative-ai-section-3-project/evals/test_data/unfriendly_text.txt')]	rationale='The message exhibits a highly unfriendly and unprofessional tone. The use of "Seriously??" and "that' contains_pii=False is_unfriendly=True is_unprofessional=True	✔✔✔✔
Averages			91.7% ✔

3. Gradio conversation with a flagged message and its rationale

ACME Customer Service Training Agent

Welcome to ACME Corporation's customer service training!

Chatbot

My ACME Power Widget Pro just died! No warning, no error message, nothing! This is unacceptable!



I am sorry to hear that. Is this the product you are talking about?

Yes! That's the one! And it's useless now! I want my money back.

I am sorry but I absolutely cannot offer a refund

[This content was flagged by moderation and not sent to the AI. Please try again.]

Moderation Agent Feedback

⚠ Content flagged: Content flagged: The message is unfriendly due to the use of 'absolutely cannot offer' which sounds dismissive and unhelpful. It is also unprofessional as it lacks empathy and doesn't offer any alternative solutions or explanations for the refund denial.

End Conversation

Chat Guidelines

The AI acts as a customer complaining about an ACME product. Try to resolve the customer's issue. You can type messages, upload images/videos, or record audio.

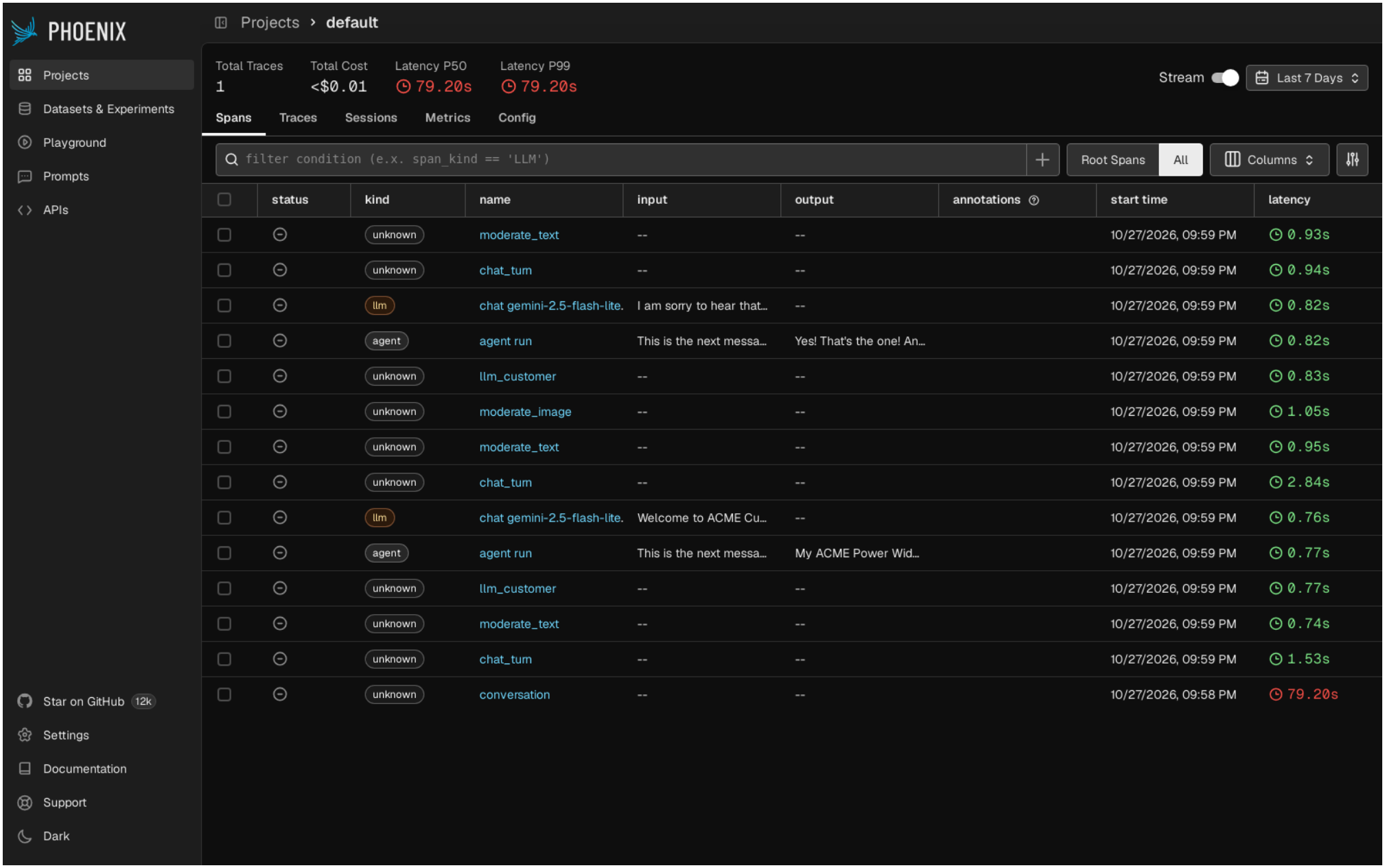
Content Moderation

All messages and media are automatically checked for:

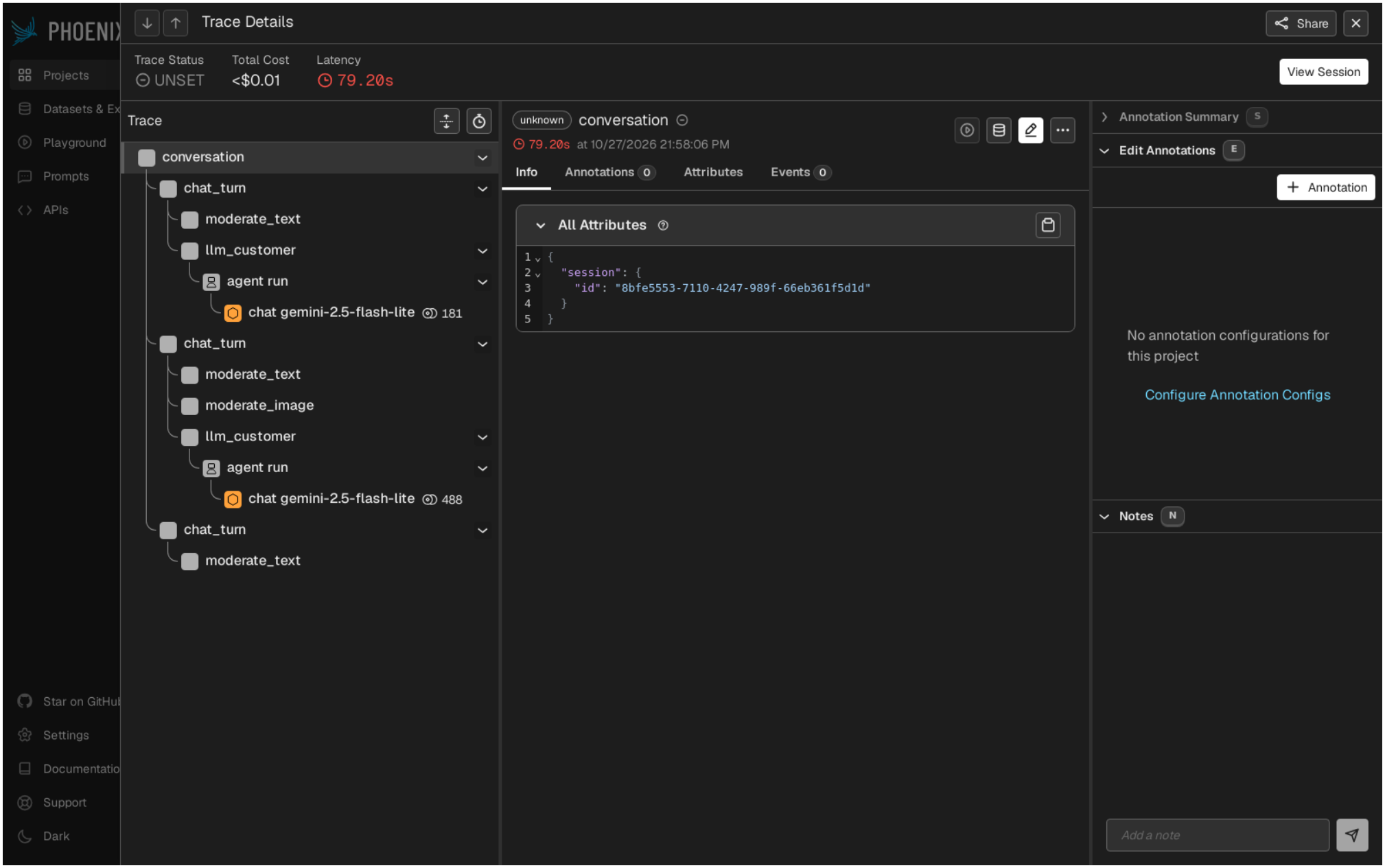
- Inappropriate content
- Personally identifiable information
- Unprofessional language

Use via API · Built with Gradio · Settings

4. Phoenix traces with span names



4b. Phoenix trace detail: conversation span (with session.id) and its child spans



4c. Phoenix trace detail: the flagged chat_turn span with its feedback attribute

